

Accounting and Accountability

Week 10



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Key concepts

Creating a flexible budget

- **static budget** – our original budget – labelled static because it does not move or change. We use our projected sales quantities and prices, materials, labour and overhead to generate our budgeted profit.
- **flexible budget** – using all of the same assumptions about sales price, cost of raw materials and cost of labour – but adjusted for the actual units sold.
- A flexible budget flexes the static budget for each anticipated level of production. This flexibility allows management to estimate what the budgeted numbers would look like at various levels of sales.
- Flexible budgets are prepared at the end of each analysis period (usually monthly), rather than in advance, since the idea is to compare the operating income to the expenses deemed appropriate at the actual production level.

Key concepts

Creating a flexible budget

- Lobster Instant Noodles make instant ramen noodles in cups that are commonly eaten by university students across the world. The business had planned to sell 25,000 units in 2022 at a price of \$6.50. There were direct variable costs for materials \$0.80, labour \$1.00, overhead \$0.50 and selling and administration \$0.50. There were also fixed costs of \$25,000 related to the factory and \$25,000 related to selling and administration.
- The static budget for Lobster Instant Noodles in 2022 is as follows:

Lobster Instant Noodles Static Budget For year ended 30 June 2022		
Budgeted sales		25,000
Budgeted price		\$ 6.50
Total sales		\$ 162,500.00
	Per unit	
Less Cost of Goods sold		
Direct material	\$ 0.80	\$ 20,000.00
Direct labour	\$ 1.00	\$ 25,000.00
Variable overhead	\$ 0.50	\$ 12,500.00
Fixed overhead		\$ 25,000.00
		\$ 82,500.00
GROSS PROFIT		\$ 80,000.00
Variable selling and admin	\$ 0.50	\$ 12,500.00
Fixed selling and admin		\$ 25,000.00
NET PROFIT		\$ 42,500.00

Key concepts

Creating a flexible budget

However, things did not go as plan in 2022. Lobster Instant Noodles faced significant issues related to their supply chain – delays in getting raw materials, employees who became unwell with COVID19 or needing to isolate resulted in performance less than what the business expected.

The table below shows that Lobster Instant Noodles sold 7000 less units and instead of making a budgeted \$42,500 of profit, instead made just \$900. But is the comparison below fair? Of course, if you sell 18,000 less units, you would expect profit to be less – this is where the flexible budget comes in.

Lobster Instant Noodles				
		Static Budget	Actual	Variance
Budgeted sales (units)		25,000	18,000	(7,000)
Budgeted price		\$ 6.50		
Total sales		\$ 162,500.00	\$ 108,000.00	\$ (54,500.00)
	Per unit			
Less Cost of Goods sold				
Direct material	\$ 0.80	\$ 20,000.00	\$ 21,600.00	\$ 1,600.00
Direct labour	\$ 1.00	\$ 25,000.00	\$ 18,000.00	\$ (7,000.00)
Variable overhead	\$ 0.50	\$ 12,500.00	\$ 8,100.00	\$ (4,400.00)
Fixed overhead		\$ 25,000.00	\$ 24,000.00	\$ (1,000.00)
		\$ 82,500.00	\$ 71,700.00	\$ (10,800.00)
GROSS PROFIT		\$ 80,000.00	\$ 36,300.00	\$ (43,700.00)
Variable selling and admin	\$ 0.50	\$ 12,500.00	\$ 9,900.00	\$ (2,600.00)
Fixed selling and admin		\$ 25,000.00	\$ 25,500.00	\$ 500.00
NET PROFIT		\$ 42,500.00	\$ 900.00	\$ (41,600.00)

To create the flexible budget – we use the standard amounts for the sales price, materials, labour, overhead, and selling and administration costs. We can then compare our static budget, flexible budget and actual.

Lobster Instant Noodles					
		Static Budget	Flexible Budget	Actual	Flexible Variance
Budgeted sales (units)		25,000	18,000	18,000	
Budgeted price		\$ 6.50	\$ 6.50		
Total sales		\$ 162,500.00	\$ 117,000.00	\$ 108,000.00	\$ (9,000.00)
	Per unit				
Less Cost of Goods sold					
Direct material	\$ 0.80	\$ 20,000.00	\$ 14,400.00	\$ 21,600.00	\$ 7,200.00
Direct labour	\$ 1.00	\$ 25,000.00	\$ 18,000.00	\$ 18,000.00	\$ -
Variable overhead	\$ 0.50	\$ 12,500.00	\$ 9,000.00	\$ 8,100.00	\$ (900.00)
Fixed overhead		\$ 25,000.00	\$ 25,000.00	\$ 24,000.00	\$ (1,000.00)
		\$ 82,500.00	\$ 66,400.00	\$ 71,700.00	\$ 5,300.00
GROSS PROFIT		\$ 80,000.00	\$ 50,600.00	\$ 36,300.00	\$ (14,300.00)
Variable selling and admin	\$ 0.50	\$ 12,500.00	\$ 9,000.00	\$ 9,900.00	\$ 900.00
Fixed selling and admin		\$ 25,000.00	\$ 25,000.00	\$ 25,500.00	\$ 500.00
NET PROFIT		\$ 42,500.00	\$ 16,600.00	\$ 900.00	\$ (15,700.00)

Quick Check

A business makes and sells candy. At the start of the year a business estimates sales revenue of \$20,100,000, with sales units of 300,000. The budget direct material cost is \$7,500,000.

The actual sales for the year were 320,000 units, with actual sales revenue of \$21,760,000. The business actually spent \$8,640,000 on direct materials.

What is the budget sales price?

1. I don't know.
2. \$62.81
3. \$68
4. \$67

Quick Check

A business makes and sells candy. At the start of the year a business estimates sales revenue of \$20,100,000, with sales units of 300,000. The budget direct material cost is \$7,500,000.

The actual sales for the year were 320,000 units, with actual sales revenue of \$21,760,000. The business actually spent \$8,640,000 on direct materials.

What is the flexible budget sales revenue?

1. \$21,760,000
2. \$20,100,000
3. \$21,440,000
4. \$20,400,000

Quick Check

A business makes and sells candy. At the start of the year a business estimates sales revenue of \$20,100,000, with sales units of 300,000. The budget direct material cost is \$7,500,000.

The actual sales for the year were 320,000 units, with actual sales revenue of \$21,760,000. The business actually spent \$8,640,000 on direct materials.

What is the budget direct material cost per unit?

1. \$28.80 per unit.
2. \$25 per unit.
3. I don't know.
4. \$27 per unit.

Quick Check

A business makes and sells candy. At the start of the year a business estimates sales revenue of \$20,100,000, with sales units of 300,000. The budget direct material cost is \$7,500,000.

The actual sales for the year were 320,000 units, with actual sales revenue of \$21,760,000. The business actually spent \$8,640,000 on direct materials.

What is the flexible budget for direct materials?

1. \$8,100,000
2. \$8,000,000
3. \$8,640,000
4. \$7,500,000

Key concepts

Conducting a variance analysis using the static and flexible budget

- ◆ The variance is the difference between the flexible budgeted performance and the actual performance. **Flexible budget variance** = actual results- flexible budget
- ◆ Identifying the variances do not necessarily tell us what is causing the issues, but highlights the issue so that management and the business can investigate the cause.

Comparing Favourable to Unfavourable Variances

Favourable

Actual Sales > Budgeted Sales

Actual Expenses < Budgeted Expenses

Unfavourable

Actual Sales < Budgeted Sales

Actual Expenses > Budgeted expenses

- It is easy to understand that an unfavourable variance may be a problem. But that is not always true, as a higher rate for wages may mean the company has a higher quality employee who is able to waste less material.
- Likewise, having a favourable variance indicates that more revenue was earned or less expenses were incurred but further analysis can indicate if costs were cut too far and better materials should have been purchased.

We can see that we spent more on direct materials, but less on manufacturing overhead. We spent more on selling and administration costs overall. Variable analysis is like a torch. It spotlights for us where the issues may be – but we must evaluate whether the variance is significant (is it big enough to warrant investigation) and then conduct further analysis.

Lobster Instant Noodles						
		Static Budget	Flexible Budget	Actual	Flexible Variance	
Budgeted sales (units)		25,000	18,000	18,000		
Budgeted price		\$ 6.50	\$ 6.50			
Total sales		\$ 162,500.00	\$ 117,000.00	\$ 108,000.00	\$ (9,000.00)	Unfavourable
	Per unit					
Less Cost of Goods sold						
Direct material	\$ 0.80	\$ 20,000.00	\$ 14,400.00	\$ 21,600.00	\$ 7,200.00	Unfavourable
Direct labour	\$ 1.00	\$ 25,000.00	\$ 18,000.00	\$ 18,000.00	\$ -	
Variable overhead	\$ 0.50	\$ 12,500.00	\$ 9,000.00	\$ 8,100.00	\$ (900.00)	Favourable
Fixed overhead		\$ 25,000.00	\$ 25,000.00	\$ 24,000.00	\$ (1,000.00)	Favourable
		\$ 82,500.00	\$ 66,400.00	\$ 71,700.00	\$ 5,300.00	Unfavourable
GROSS PROFIT		\$ 80,000.00	\$ 50,600.00	\$ 36,300.00	\$ (14,300.00)	Unfavourable
Variable selling and admin	\$ 0.50	\$ 12,500.00	\$ 9,000.00	\$ 9,900.00	\$ 900.00	Unfavourable
Fixed selling and admin		\$ 25,000.00	\$ 25,000.00	\$ 25,500.00	\$ 500.00	Unfavourable
NET PROFIT		\$ 42,500.00	\$ 16,600.00	\$ 900.00	\$ (15,700.00)	Unfavourable

Quick Check

A business makes and sells candy.

At the start of the year a business estimates sales revenue of \$20,100,000, with sales units of 300,000. The actual sales for the year were 320,000 units, with actual sales revenue of \$21,760,000.

The flexible budget for sales is $320,000 \times \$67 = \$21,440,000$.

What is the sales revenue flexible variance?

1. \$1,340,000
2. \$320,000
3. \$1,660,000
4. I don't know.

Quick Check

When actual sales are more than flexible budget sales, this variance is called:

1. Favourable
2. Unfavourable

Quick Check

A business makes and sells candy.

At the start of the year a business estimates the direct material cost will be \$7,500,000 (This is the static budget amount). The business actually spent \$8,640,000 on direct materials.

The direct material flexible budget cost is calculated using actual sales of 320,000 x budget direct material cost per unit \$25 = \$8,000,000.

What is the flexible budget variance for direct materials?

1. \$\$1,140,000
2. \$640,000
3. I don't know.
4. \$500,000

Quick Check

When the actual expenses are more than the flexible budget variances this is called:

1. Unfavourable.
2. Favourable.

Key concepts

Identifying the causes of variances

- Sales-related variances

Our budgeted sales is calculated by multiplying our predicted sales (developed by marketing and management) by the sales price.

$$\text{Budgeted sales} = \text{Predicted sales (units)} \times \text{Sales price}$$

Our flexible budget uses the same units sold as the actual sales for the year. This means that the only other factor that could result in actual performance being different from budgeted performance is the **Sales Price**.

If we work backwards, \$108,000 divided by 18,000 units provides us with an Actual Sales Price of \$6.00. This is \$0.50 less than our original budget of \$6.50.

This could include reasons such as (but not limited to):

- The business was losing customers to a competitor who was charging less than \$6.50
- The economic impact of COVID19 meant that customers were not willing to pay the original budgeted price

Key concepts

Identifying the causes of variances

Variance	What caused the variance?	What business decisions need to be made?
Sales	Sales price is higher or lower than expected. Brainstorm why the price changed. Here are some things to consider: • Competitor's sales prices. • Is there a new competitor? • Impact of COVID-19 or other current events. • Economic situation. • Did the price change in response to market research? • Are we trying to market our product as a more 'luxury' product with a higher price? • Have our costs increased, and we are passing them on to customers? • Have we been discounting or offering sales to customers to maintain our planned level of sales units?	Is this a permanent change or a one-off? Do we need to plan for a change in sales price for the next budget? Do we need to spend more on marketing?

Key concepts

Identifying the causes of variances

- **Material-related variances**

When preparing the budget for our direct materials, we take our predicted sales and multiply this by the budgeted materials used per unit and the price of raw materials.

$$\begin{array}{ccccccc} \text{Direct} & & & & \text{Budgeted} & & \\ \text{materials} & = & \text{Sales} & & \text{materials used} & \times & \\ \text{budget} & & \text{(units)} & & \text{per unit} & & \text{Price of raw} \\ & & & & & & \text{materials} \end{array}$$

Lobster Instant Noodles budgeted for \$0.80 per unit of raw materials. However, the business actually used \$1.20 of raw materials (\$21,600 in total raw materials divided by 18,000 units = \$1.20 per unit). Presume we are given the following new information

- the raw materials were priced at \$0.40 per 100g and each cup should use 200g of raw materials
- raw material prices did not change since the budget was prepared. Then we can deduce that the business must have used 300g of raw materials to make each cup of noodles

Why did the business use more raw materials?

- the raw materials were of poorer quality and there was more wastage
- the production process had become more inefficient

Key concepts

Identifying the causes of variances

Variance	What caused the variance?	What business decisions need to be made?
Direct materials	Did we use more or less quantity of materials than planned? • Did we buy a poor quality material that was harder to use, and we had more waste? • Did we use the materials efficiently in the factory? • Did we change suppliers? • Were there more mistakes made in the factory? • Did we buy a new machine that used the materials more efficiently? Did we pay more or less to buy the materials than planned? • Did we buy more expensive or cheaper materials? • Did the price of our materials change due to more overall demand for the materials or less supply or a combination of both?	Should we find a new supplier to buy a better quality or cheaper material? Could we get a cheaper price if we buy in bulk? Review the production process. Can we use the materials more efficiently?

Key concepts

Identifying the causes of variances

- Labour-related variances

When preparing the budget for our direct labour, we take our predicted sales and multiply this by the budgeted labour hours used per unit and the rate of pay for our labour / employees.

$$\text{Direct labour budget} = \text{Sales (units)} \times \text{Budgeted labour hours used per unit} \times \text{Rate of pay for labour}$$

In our Lobster Instant Noodles example, we haven't been provided with this information so far. However, if we compare our actual and flexible budget for direct labour – you'll notice that nothing has changed. There is 0 variance. What does this mean?

It could mean one of a few options:

- That the business used the expected number of labour hours per unit and the rate of pay is as expected
- We could be paying employees more per hour, but they may be more efficient – resulting in less hours but at a higher rate, and the same total wage bill
- We could be paying employees less per hour, but they may be inefficient – resulting in greater hours, but at a lesser rate, and the same total wage bill.

Key concepts

Identifying the causes of variances

Variance	What caused the variance?	What business decisions need to be made?
Direct labour	Did our employees work slower or faster than expected? • How experienced and skilled are our employees? Did we pay employees more or less per hour than expected? • Is labour scarce? • Do we prefer to use more experienced and skilled labour, which costs more? • Did government regulation require us to increase the amount paid?	What will be more profitable for our business: more experienced, qualified labour that costs more or less experienced, cheaper labour? When the budget was created, did it include some budget slack to make it easier for employees to get a favourable result (this could apply to all these areas)?

Key concepts

Identifying the causes of variances

Overhead and selling and administration-related variances

The overhead costing process is quite complicated and we have not gone into detail as to how overhead is calculated. Therefore, we do not expect students to be able to conduct variance analysis in any great detail for overhead and selling and administration-related variances. Simply understanding favourable and unfavourable and being able to brainstorm potential reasons for overall changes is sufficient.

Overhead variances	Selling and administration variances
• The business negotiated a decrease in a fixed cost such as the rent for the factory or equipment • The business installed solar panels on the factory roof resulting in less paid for electricity to run the manufacturing equipment • The business re-evaluated how they calculated their overhead costs and found an error resulting in a change in the overhead	• The business changed advertising strategies from fixed costs in traditional media (print, television, radio) to a greater focus on social media spending which has a greater variable component • The business shifted from standard post services to express post to keep customers happy and encourage sales

Key concepts

Describe how companies use variance analysis

- The starting point is the determination of standards against which to compare actual results.
- Many companies produce variance reports, and the management responsible for the variances must explain any variances outside of a certain range.
- Some companies only require that unfavorable variances be explained, while many companies require both favourable and unfavourable variances to be explained.
- The proper use of variance analysis is a significant tool for an organisation to reach its long-term goals. When its accounting system recognises a variance, a business needs to understand the significant influence of accounting not only in recording its financial results, but also in how reacting to that variance can shape management's behaviour toward reaching its goals.

Key concepts

Standard costs

- Management can use standard costs to prepare the budget for the upcoming period, using the past information to possibly make changes to production elements.
- **Standard costs** are a measurement tool and can thus be used to evaluate performance.
- As you've learned, management may manage "to the variances" and can manipulate results to meet expectations. To reduce this possibility, performance should be measured on multiple outcomes, not simply on standard cost variances.

Standard Costs

Pros

- Useful when developing a future budget
- Can be used as a benchmark for performance and quality expectations
- Can individually identify areas of success and areas for improvement

Cons

- Might ignore customer and employee satisfaction rates
- Information could be historical data and not useful in real-time decision-making needs
- The system to manage and develop standard costs requires a lot of resources, which could be costly and time consuming

Tutorial activities--Tutorial 10A

A business prepared a budget with sales of 15,000 units at \$10 each. The business actually sold 12,000 units and earned sales revenue of \$132,000. What is the flexible budget amount for sales revenue?

- a.\$132,000
- b.\$120,000
- c.\$150,000
- d.I don't know.

What is a flexible budget variance?

- a.The difference between the actual results and flexible budget.
- b.When the business spends more than it planned to.
- c.When a business uses more resources than it planned to use.
- d.I don't know.

Tutorial activities--Tutorial 10A

A business prepared a budget with sales of 15,000 units at \$10 each. The business actually sold 12,000 units and earned sales revenue of \$132,000. What is the flexible budget variance for sales revenue?

- a.\$30,000
- b.\$18,000
- c.\$12,000
- d.I don't know.

A business prepared a budget with sales of 15,000 units at \$10 each. The business actually sold 12,000 units and earned sales revenue of \$132,000. Is the flexible budget variance favourable or unfavourable?

- a.Favourable
- b.Unfavourable
- c.I don't know.

Tutorial activities--Tutorial 10A

8. A business prepared a budget with sales of 15,000 units at \$10 each. The business actually sold 12,000 units and earned sales revenue of \$132,000. What has likely caused the favourable flexible budget variance for sales?

- a. The business sold more units.
- b. The cost of materials was cheaper.
- c. The selling price per unit was higher than planned.
- d. I don't know.

Tutorial activities--Tutorial 10A variance analysis

Complete variance analysis for Cherry for the next year using the information below and then answer the questions.

Budget

Based on the past variance analysis, this is Cherry's budget for the next year:

- Budget sales price: \$55. Due to global inflation and an increase in all her budgeted costs (see below), Cherry decided to pass some of these costs onto the customer and increase the sales price.
- Budget units: 5,500. Cherry believes with the easing restrictions that her tops will grow in popularity and she will sell more unit in the next year.
- All the budget costs are expected to increase due to global inflation. Direct material costs will increase to \$17 per unit.
- Direct labour costs are budgeted to decrease to \$18 per unit.
- Variable overhead \$0.20 per unit.
- Fixed overhead \$5,500.
- Variable selling and administration \$0.60 per unit.
- Fixed selling and administration \$21,000.

Actual results

Cherry gets the following information from her accounting software MYOB:

- Sales units: 6,000.
- Sales revenue: \$300,000
- Direct materials cost: \$100,000
- Direct labour cost: \$110,000
- Variable overhead: \$600
- Fixed overhead: \$6,000
- Variable selling and administration: \$3,000
- Fixed selling and administration: \$22,00

Tutorial activities--Tutorial 10A variance analysis

Cherry Love For the year ended 30 June						
		Static budget	Flexible budget	Actual	Flexible budget Variance	Favourable or unfavourable
Budget sales units						
Budget sales price						
Total budget sales						
	Budget per unit					
<i>Less: Cost of goods sold</i>						
Direct materials						
Direct labour						
Variable overhead						
Fixed overhead						
Gross profit						
Variable selling and admin						
Fixed selling and admin						
Net profit						

1. What was the actual sales price?
2. What caused these variances? Focus on the sales, materials and labour.
3. What should Cherry do?

Tutorial activities--Tutorial 10B

Complete variance analysis for Marketing Love ♥ using the information below and then answer the questions.

Budget

- Budget sales price: \$100
- Budget units: 4,200
- Direct material cost per unit: \$24
- Direct labour costs per unit: \$6
- Variable overhead \$6 per unit
- Fixed overhead \$21,000
- Variable selling and administration \$1 per unit.
- Fixed selling and administration \$3,400.

Actual results

Raphael gets the following information from the accounting software MYOB for the year:

- Sales units: 5,000
- Sales revenue: \$550,000
- Direct materials cost: \$129,000
- Direct labour cost: \$25,000
- Variable overhead: \$32,000
- Fixed overhead: \$22,000
- Variable selling and administration: \$6,000
- Fixed selling and administration: \$7,000

Tutorial activities--Tutorial 10B

Write good quality answers to these questions:

1. What was the actual sales price?
2. What caused these variances? Focus on the sales, materials and labour.
3. What should Marketing Love ❤️ do to keep growing the business in the future?

Marketing Love ❤️						
For the year ended 30 June						
		Static budget	Flexible budget	Actual	Flexible budget variance	Favourable or unfavourable
Budget sales units						
Budget sales price						
Total budget sales						
	Budget per unit					
<i>Less: Cost of goods sold</i>						
Direct materials						
Direct labour						
Variable overhead						
Fixed overhead						
Gross profit						
Variable selling and admin						
Fixed selling and admin						
Net profit						

Tutorial problems week 10

Follow the guide of budget assumptions in the previous **Marketing Love** ❤️ case, design your own business, prepare a variance analysis for the this year and present your findings.

- You will work in groups (4-5 in one group) on tutorial problems.
- You will prepare a PPT and present the solution with the class in tutorial week.
- All group members need to participate and explain the solution and also answer any further questions from the class.
- Time limit: 15-20 minutes.